AN INVESTIGATION INTO

Modulations

Color, Music, and Time

Every color is a frequency and every musical note is a frequency. Carry a note upward octave by octave and it becomes light. This is an account of what that mapping reveals across the twelve chromatic pitches — and of whether a palette can move over time the way a piece of music changes key.

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PART I

The Method

Following the Octave

The starting point is a single physical principle: the octave. In music, an octave is an exact doubling of frequency. A4 is 440 Hz. A5 is 880 Hz. They are perceived as the same note at different registers — and this perception is grounded in physics: the overtones of 880 Hz align perfectly with those of 440 Hz. The 2:1 ratio is the most fundamental consonance in acoustics.

Light is also a frequency. The question is straightforward: if you keep doubling a musical note's frequency — octave by octave — how many doublings does it take to reach visible light? And when you get there, where do you land?

> **A note of caution:** Octave identity — the perception that 440 Hz and 880 Hz are "the same note" — is a property of the human auditory system, not a universal physical law. And the auditory system itself is contingent: a different atmosphere would differentially absorb and refract different frequencies, meaning music would arrive at the ear sounding different even from the same source. Different cochlear anatomy — larger ears, as in elephants, or smaller, as in bats — would shift the entire range of what is heard as pitch and what is felt as vibration. Our music theory is not a description of frequency relationships in the abstract; it is a description of how *this* auditory system, evolved in *this* atmosphere, responds to those relationships.
>
> The same is true for color. The visible spectrum is physically continuous, but how it is carved into colors depends on the three cone types in the human retina — and those vary measurably between individuals. People with anomalous trichromacy see two wavelengths as identical that others distinguish. Tetrachromats — a significant minority of women who possess a fourth cone type — may distinguish colors invisible to most people, though estimates of their prevalence vary widely across studies. What we call "color" is not a property of light alone. It is the product of three things: the light itself, the biological hardware receiving it, and the cultural vocabulary that taught us which distinctions matter.
>
> A photon at 575 THz therefore has no physical relationship to a 261 Hz sound wave simply because one is 2^{41} times the other. They are different physical phenomena, processed by different biological systems, both shaped by the same evolutionary environment. The octave relationship is mathematically real; whether it is perceptually or physically meaningful across domains is a question this investigation cannot fully resolve. That uncertainty is not a failure of the framework. It is the honest boundary of what the mathematics, by itself, can tell us.

Finding the Anchor

Before asking how many doublings it takes to reach visible light, we need to ask a more precise question: where in the audible range should we start?

The visible light spectrum runs from approximately 428 THz (700nm, deep red) at one end to 789 THz (380nm, violet) at the other. When we apply the $\times 2^{41}$ octave transformation and ask which musical note lands at the *bottom* of the visible spectrum — at the deep red boundary — the answer is specific: **G3 (196 Hz)**.

G3 $\times 2^{41}$ = 431 THz = 696nm — deep red, right at the red boundary of visible light.

This is not an arbitrary choice. G3 is the note whose octave transformation lands at the beginning of the visible spectrum. It is the natural anchor in the same way that the red end of the spectrum is the natural beginning point when discussing visible light: both are the physically determined starting edge of their respective ranges.

Starting from 261.63 Hz (Middle C) and doubling repeatedly, for comparison:

C4	261.63 Hz	Audible sound
C20	17.2 MHz	Radio
C30	17.6 GHz	Microwave
C40	18.1 THz	Infrared
C41	575 THz	Visible light (green)

Middle C $\times 2^{41}$ lands at 521nm — green, roughly in the centre of the visible band. This tells us something real: Middle C, the cultural and acoustic reference point of Western music, lands in the middle of the solar-relevant spectral region (more on this in Part X). But it is not the bottom of the visible spectrum. G3 is.

> **Note on tuning conventions:** These calculations use A4 = 440 Hz (ISO 16:1975). Different tuning standards shift all wavelengths slightly. This has been tested across eleven historical tuning standards from Baroque A415 to modern A444. The results described below are robust across all of them, with one precisely characterised exception noted where relevant.

All Twelve Notes

Starting from G3 (196 Hz) and applying the $\times 2^{41}$ transformation to every note of the chromatic scale produces the following map:

NOTE	HZ	THZ (×2⁴¹)	WAVELENGTH	ACCURATE COLOR	VISIBLE?
G	196.00	431.01	696 nm	deep red	YES
G♯	207.65	456.64	657 nm	red	YES
A	220.00	483.79	620 nm	vermilion (red-orange)	YES
A♯	233.08	512.56	585 nm	amber-yellow	YES
B	246.94	543.04	552 nm	harlequin (yellow-green)	YES
C	261.63	575.33	521 nm	green	YES
C♯	277.19	609.54	492 nm	cyan-blue	YES
D	293.67	645.79	465 nm	blue	YES
D♯	311.13	684.19	439 nm	indigo	YES
E	329.63	724.87	414 nm	violet	YES
F	349.23	767.97	391 nm	deep violet	YES
F♯	369.99	813.63	369 nm	near-UV threshold	no

Eleven of the twelve chromatic notes land in the visible spectrum. The remaining note — F♯, the tritone — falls 11.3nm past the violet boundary into near-ultraviolet.

This is the maximum number of chromatic notes that can ever fit in the visible band, regardless of starting pitch. The visible spectrum spans 10.58 semitones (0.882 octaves of light). The chromatic scale has 12 semitones per octave. No starting pitch can fit all 12 — 1.42 semitones will always overshoot.

> **A note of caution — what "11 of 12" does and doesn't show:** It is tempting to read "11 of 12 notes fit" as evidence that G3 is a special or improbable anchor. It isn't, and the investigation should say so plainly. Because the visible window (10.58 semitones) is only 1.42 semitones short of a full octave, landing on 11 visible notes is the *typical* outcome for almost any anchor — the geometry makes it hard *not* to get 11. Every historical tuning standard this project tested, from Baroque A415 to modern A444, produces 11 visible notes; that consistency is a symptom of how forgiving the geometry is, not proof that G3 was singled out by the physics. What the geometry does *not* determine in advance is *which* note gets excluded. Under the specific, independently-fixed A440 tuning this project uses — a convention chosen for reasons that have nothing to do with color — the excluded note happens to be F♯, the tritone. That narrower fact, not the "11 of 12" headline, is the part of this finding that isn't guaranteed by arithmetic alone. See Claim 1.12 in the Claims Register.
>
> **A note of caution:** Color names are not standardised in the way musical note frequencies are. A4 = 440 Hz is fixed by international agreement (ISO 16:1975). No equivalent body has fixed “blue” to a specific wavelength — color names cover broad, culturally-defined bands. The wavelengths in the table above are precise; the color names are the closest accurate spectral descriptions for those wavelengths. The implications of this asymmetry are discussed in Part VIII.

Sources for Part I: ISO 16:1975 (tuning standard); standard optics references for visible spectrum boundaries.

PART II

The Discovery

An Almost-Complete Chromatic Scale

Eleven of the twelve chromatic notes land in the visible spectrum, running in order from deep red to deep violet. This is not a random selection. Look at what it contains — and what it excludes.

The note that falls outside the visible spectrum is **F♯** — the tritone above C. The tritone is music theory's most dissonant interval. Medieval theorists called it the *diabolus in musica* — the devil in music. It was avoided in sacred composition for centuries. It sits at the exact midpoint of the octave, equidistant from both the tonic and its octave above — the most harmonically unstable position possible. At 369nm, it sits 11.3nm past the violet boundary of visible light.

The visible spectrum, under the octave-doubling transformation, is an almost-complete chromatic scale. The one missing note is the most dissonant interval in Western music.

This gap is physically determined. The visible spectrum spans 10.58 equal-tempered semitones — just 1.42 semitones short of a full chromatic octave. That shortfall is fixed by the physics of the atmospheric optical window and the mathematics of equal temperament. No choice of anchor can close it. The tritone happens to be the note that falls in that gap, determined by where the atmosphere ends the visible transmission window (~380nm) relative to the equal-tempered semitone spacing above red.

> **A note of caution:** The UV boundary is not a sharp line. Individual variation in UV sensitivity exists; some people can detect light to approximately 360nm. F♯ at 369nm is comfortably past the 380nm standard boundary. The significance of this result was noticed after the calculation, not predicted before it — which means it should be treated as a suggestive observation rather than a confirmed finding.

The Warm and Cool Registers

The 11-note visible chromatic scale divides naturally into two registers:

The warm register (G, G♯, A, A♯, B): Red, deep red, vermilion, amber-yellow, harlequin. These are the colors everyone intuitively recognises as warm — the colors of fire, heat, blood, late afternoon light.

The cool register (C, C♯, D, D♯, E, F): Green, cyan-blue, blue, indigo, violet, deep violet. These are the colors of water, shadow, sky, distance.

The boundary between them falls between B (552nm, harlequin) and C (521nm, green). These are two different ways of describing what that boundary is, and both are correct — they are answering different questions.

Where in the spectrum does warm become cool? The spectral answer is: between B and C, a gap of approximately 31nm. The interval between those two adjacent notes is a **semitone** — the smallest step in the chromatic scale.

What is the harmonic significance of that divide? The structural answer is: the divide is between the anchor (G) and the tonic (C) — a **perfect fifth**. G3 is the note the entire mapping is built on; C4 is the note five steps above it. The warm register contains all the notes from the anchor up to and including B. The cool register begins at C. In scale terms, the boundary is not the B→C semitone but the G→C relationship underlying the whole arrangement — the dominant-to-tonic interval, the most fundamental consonance in Western music after the octave.

The semitone is where the boundary *sits* in the spectrum. The perfect fifth is what the boundary *means* harmonically. The warm half of the visible spectrum corresponds to the notes from the anchor to the dominant. The cool half corresponds to the notes from the tonic upward.

This is not constructed. It follows directly from G3 being the anchor.

The Tetrachord Within the Scale

The four notes C, D, E, F constitute the Greek diatonic tetrachord — the oldest and most foundational structure in the history of Western and Near Eastern music theory. The word comes from Greek: *tetra* (four) and *chorde* (string). Before the time of Terpander (c. 710 BC), the Greek lyre had four strings. The outer two — tonic and perfect fourth — were fixed. The two inner notes were movable, and their tuning defined the character of the music.

Under the corrected mapping, C, D, E, F all land in the cool upper register of the visible chromatic scale — green through deep violet. The tetrachord is not the full story; it is the upper portion of the cool half of an 11-note visible scale. But it is fully visible, and it occupies a structurally coherent position: the four notes bounded by the perfect fourth (C to F), sitting at the top of the visible range with F at the deep-violet threshold.

The diatonic tetrachord (C, D, E, F) is joined by two additional chromatic notes in the visible band: C♯ and D♯. These are not incidental — they are precisely the notes the Greeks used when constructing the other two genera of their system. In the chromatic genus, the movable inner notes descended to D♯. The visible spectrum contains not just the diatonic tetrachord but the full set of notes from which the Greeks derived all three genera.

> **A note of caution:** The Greek tetrachord was defined by its interval structure, not its absolute pitch. The Greeks had no concept of a fixed frequency for any note. Calling the visible notes “the Greek tetrachord” requires the qualification that what matches is the interval pattern and note count, not a direct frequency correspondence with ancient Greek practice.

The Eleven Visible Notes in Detail

G — Deep Red (696nm) The anchor note. G3 × 2^{41} = 696nm — the bottom edge of the visible spectrum, the deepest red. In music, G is the dominant — the note with the strongest gravitational pull back toward C. The anchor of the visible scale is the note most urgently directed toward its tonic.

G♯ — Red (657nm) A semitone above the anchor, landing in pure red. G♯ is the chromatic alteration of the anchor — a slight sharpening that shifts deep red into full red.

A — Vermilion (620nm) A whole tone above G, landing in vermilion — the precise spectral name for red-orange. In music, A is the universal reference pitch. A4 = 440 Hz is the international tuning standard, the note to which every instrument in the world is calibrated. The universal musical reference sits in vermilion.

A♯ — Amber-Yellow (585nm) The minor second above A, landing in amber-yellow. This is the transition from the red-orange family into the yellow register — warm, luminous, the color of late-afternoon light and ripe grain.

B — Harlequin (552nm) The major second above A, landing in harlequin — the spectral color precisely at the yellow-green transition, between chartreuse and green. B is the leading tone: one semitone below C, with the most urgent directional pull in Western harmony toward the tonic above it. The warm register's highest note, directed urgently toward the cool.

C — Green (521nm) The tonic. Middle C × 2^{41} = 521nm — green, near the centre of the visible spectrum, within the solar-relevant spectral region. The tonic is green. The boundary between the warm and cool registers of the visible scale.

C♯ — Cyan-Blue (492nm) The minor second above the tonic, landing in cyan-blue — the transitional color between cyan and blue. In Greek music theory, C♯ occupies the inter-genus tonal space: it falls between the diatonic inner note (D) and the chromatic inner note (D♯), the position the Greeks moved differently across their genera. It is present in the visible band; it is not the defining note of any of the three standard genera.

D — Blue (465nm) A whole tone above C, landing in true blue. The major second is an open, unresolved interval; it connects but does not settle. The step from 521nm to 465nm crosses from the warm-adjacent green into the clearly cool blue range.

D♯ — Indigo (439nm) The minor third above C, landing in indigo — the transitional band between blue and violet. D♯ is the movable inner note of the chromatic genus — the defining note that distinguishes the chromatic tetrachord (C, D♯, E, F) from the diatonic (C, D, E, F). In Western tonal music, the minor third gives minor chords their quality of shadow.

E — Violet (414nm) The major third above C, landing at the extreme violet edge of human vision. The major third defines whether a chord sounds major or minor — it is the interval that distinguishes light from shadow in harmony. The C major chord (C-E-G) maps to green, violet, and the anchor deep red — a triad spanning the full visible scale from near-centre to extremes.

F — Deep Violet (391nm) The perfect fourth above C — one of the three primary consonances identified in acoustics (octave, fifth, fourth) — lands at the violet threshold of human vision. This is the highest note of the visible scale before the world goes ultraviolet. The perfect fourth in music occupies a similar liminal position: consonant but open, a boundary rather than a destination.

F♯ — Near-UV Threshold (369nm) The tritone — 11.3nm past the visible boundary. Displayed in the color map as the deepest perceptible violet, labeled "near-UV threshold." It is present at the edge. Past the standard boundary of human vision. It is the *diabolus in musica*, and it lives just past the threshold of sight.

Sources for Part II: Aristoxenus, Elementa Harmonica (Greek music theory and genera); standard music history for Terpander and the four-string lyre; arithmetic from Claims Register Claims 1.3, 1.5, 1.6.

PART III

The Chromatic Threshold

F♯ and the Tritone

F♯ — the tritone above C — lands at 369nm under modern tuning. The conventional visible spectrum ends at roughly 380nm. F♯ falls 11.3nm into UV-A.

Under the corrected mapping, F♯ is not simply one of several notes that miss the visible spectrum. It is the *only* note that misses. Every other chromatic note lands in the visible band. The tritone alone sits at the threshold of vision.

The tritone is music theory's most dissonant interval. Medieval theorists called it the *diabolus in musica* — the devil in music. It was avoided in sacred composition for centuries. It sits at the exact midpoint of the octave, equidistant from both the tonic and its octave above — the most harmonically unstable position possible. And it is the sole chromatic note that falls past the boundary of visibility.

The 11.3nm gap is physically determined. Changing which octave of F♯ you use does not help: all octaves of F♯ produce the same wavelength, because octave-shifting the note and adjusting the power of 2 always yield identical products. The only way to move F♯ into the visible range is to change the tuning standard. Under Baroque A415, F♯ lands at 391nm — just inside the visible boundary at the deepest violet, still at the extreme edge of human color perception. Under all post-Baroque tuning, F♯ is in the UV.

In the color map, F♯ is rendered as the deepest perceptible violet — clamped to 380nm, labeled "near-UV threshold." It is present at the edge. Present and not quite visible. The *diabolus in musica* lives at the boundary of the seeable world.

A Structural Observation

The visible spectrum, under the octave transformation, is an almost-complete chromatic scale. The one note excluded is precisely the one that music theory identifies as the boundary between consonance and dissonance — the interval that sits farthest from harmonic stability. Whether this correspondence is a profound structural truth or an elegant coincidence determined by atmospheric physics and equal temperament is a question this investigation cannot answer. Both possibilities deserve taking seriously.

> **A note of caution — this observation is narrower than it may sound:** Some note was always going to be excluded; the geometry of a 10.58-semitone window against a 12-semitone scale guarantees it (see Part II). That part is not surprising and should not be presented as if it were. What is genuinely not guaranteed by the geometry is *which* note the standard A440 tuning convention happens to exclude. That it is F♯ — the tritone, the note Western theory already singles out as maximally unstable — is a specific, checkable fact about where 440 Hz sits relative to 380–700nm, not a generic property of "almost fitting" a 12-note scale into an 11-note window. This is the correct scope of the claim: a coincidence about one particular tuning standard and one particular note, not a demonstration that the visible spectrum "wants" to be a chromatic scale.

Sources for Part III: Standard music theory for the tritone and diabolus in musica; Claims Register Claims 1.5, 1.6 for arithmetic.

PART IV

Full Chromatic Reference

Ten octaves of the chromatic scale, each note carried into visible light by the octave transform. Hue encodes pitch class; lightness encodes register. G and G♯ both render as deep red — above ~645nm the eye stops discriminating hue.

	C	C#	D	D#	E	F	F#	G	G#	A	A#	B
Oct 9	C	C#	D	D#	E	F	F#	G	G#	A	A#	B
Oct 8	C	C#	D	D#	E	F	F#	G	G#	A	A#	B
Oct 7	C	C#	D	D#	E	F	F#	G	G#	A	A#	B
Oct 6	C	C#	D	D#	E	F	F#	G	G#	A	A#	B
Oct 5	C	C#	D	D#	E	F	F#	G	G#	A	A#	B
Oct 4	C	C#	D	D#	E	F	F#	G	G#	A	A#	B
Oct 3	C	C#	D	D#	E	F	F#	G	G#	A	A#	B
Oct 2	C	C#	D	D#	E	F	F#	G	G#	A	A#	B
Oct 1	C	C#	D	D#	E	F	F#	G	G#	A	A#	B
Oct 0	C	C#	D	D#	E	F	F#	G	G#	A	A#	B

PART V

Scales

Every major and natural-minor scale, each degree shown as its mapped color. The shape of a scale becomes a sequence of hues.

MAJOR SCALES

C Major

C	D	E	F	G	A	B	C
Green	Blue	Violet	Deep Violet	Deep Red	Vermilion	Harlequin	Green

D Major

D	E	F#	G	A	B	C#	D
Blue	Violet	Near-UV	Deep Red	Vermilion	Harlequin	Cyan-Blue	Blue

E Major

E	F#	G#	A	B	C#	D#	E
Violet	Near-UV	Red	Vermilion	Harlequin	Cyan-Blue	Indigo	Violet

F Major

F	G	A	A#	C	D	E	F
Deep Violet	Deep Red	Vermilion	Amber-Yellow	Green	Blue	Violet	Deep Violet

G Major

G	A	B	C	D	E	F#	G
Deep Red	Vermilion	Harlequin	Green	Blue	Violet	Near-UV	Deep Red

A Major

A	B	C#	D	E	F#	G#	A
Vermilion	Harlequin	Cyan-Blue	Blue	Violet	Near-UV	Red	Vermilion

B Major

B	C#	D#	E	F#	G#	A#	B
Harlequin	Cyan-Blue	Indigo	Violet	Near-UV	Red	Amber-Yellow	Harlequin

NATURAL MINOR SCALES

C Natural minor

C	D	D#	F	G	G#	A#	C
Green	Blue	Indigo	Deep Violet	Deep Red	Red	Amber-Yellow	Green

D Natural minor

D	E	F	G	A	A#	C	D
Blue	Violet	Deep Violet	Deep Red	Vermilion	Amber-Yellow	Green	Blue

E Natural minor

E	F#	G	A	B	C	D	E
Violet	Near-UV	Deep Red	Vermilion	Harlequin	Green	Blue	Violet

F Natural minor

F	G	G#	A#	C	C#	D#	F
Deep Violet	Deep Red	Red	Amber-Yellow	Green	Cyan-Blue	Indigo	Deep Violet

G Natural minor

G	A	A#	C	D	D#	F	G
Deep Red	Vermilion	Amber-Yellow	Green	Blue	Indigo	Deep Violet	Deep Red

A Natural minor

A	B	C	D	E	F	G	A
Vermilion	Harlequin	Green	Blue	Violet	Deep Violet	Deep Red	Vermilion

B Natural minor

B	C#	D	E	F#	G	A	B
Harlequin	Cyan-Blue	Blue	Violet	Near-UV	Deep Red	Vermilion	Harlequin

PART VI

Chords — 60/30/10

Each triad drawn as a proportional color bar — tonic 60%, fifth 30%, third 10%. A chord becomes a palette; the ratios are the ones a designer would reach for.

C maj
MAJOR TRIAD

C	E	G
Green	Violet	Deep Red
TONIC	FIFTH	THIRD

60 · 30 · 10 (tonic/fifth/third)

D min
MINOR TRIAD

D	F	A
Blue	Deep Violet	Vermilion
TONIC	FIFTH	THIRD

60 · 30 · 10 (tonic/fifth/third)

E min
MINOR TRIAD

E	G	B
Violet	Deep Red	Harlequin
TONIC	FIFTH	THIRD

60 · 30 · 10 (tonic/fifth/third)

F maj
MAJOR TRIAD

F	A	C
Deep Violet	Vermilion	Green
TONIC	FIFTH	THIRD

60 · 30 · 10 (tonic/fifth/third)

G maj
MAJOR TRIAD

G	B	D
Deep Red	Harlequin	Blue
TONIC	FIFTH	THIRD

60 · 30 · 10 (tonic/fifth/third)

A min
MINOR TRIAD

A	C	E
Vermilion	Green	Violet
TONIC	FIFTH	THIRD

60 · 30 · 10 (tonic/fifth/third)

B dim
DIMINISHED

B	D	F
Harlequin	Blue	Deep Violet
TONIC	FIFTH	THIRD

60 · 30 · 10 (tonic/fifth/third)

G7
DOM. 7TH

G	B	D	F
Deep Red	Harlequin	Blue	Deep Violet
TONIC	FIFTH	SEVENTH	THIRD

50 · 25 · 15 · 10 (tonic/fifth/seventh/third)

Cmaj7
MAJOR 7TH

C	E	G	B
Green	Violet	Deep Red	Harlequin
TONIC	FIFTH	SEVENTH	THIRD

50 · 25 · 15 · 10 (tonic/fifth/seventh/third)

Fmaj7
MAJOR 7TH

F	A	C	E
Deep Violet	Vermilion	Green	Violet
TONIC	FIFTH	SEVENTH	THIRD

50 · 25 · 15 · 10 (tonic/fifth/seventh/third)

Am7

MINOR 7TH

A	C	E	G
Vermilion	Green	Violet	Deep Red
TONIC	FIFTH	SEVENTH	THIRD

50 · 25 · 15 · 10 (tonic/fifth/seventh/third)

Dm7

MINOR 7TH

D	F	A	C
Blue	Deep Violet	Vermilion	Green
TONIC	FIFTH	SEVENTH	THIRD

50 · 25 · 15 · 10 (tonic/fifth/seventh/third)

E7

DOM. 7TH

E	G#	B	D
Violet	Red	Harlequin	Blue
TONIC	FIFTH	SEVENTH	THIRD

50 · 25 · 15 · 10 (tonic/fifth/seventh/third)

B♭maj

MAJOR TRIAD

A#	D	F
Amber-Yellow	Blue	Deep Violet
TONIC	FIFTH	THIRD

60 · 30 · 10 (tonic/fifth/third)

C♯min

MINOR TRIAD

C#	E	G#
Cyan-Blue	Violet	Red
TONIC	FIFTH	THIRD

60 · 30 · 10 (tonic/fifth/third)

A♭maj

MAJOR TRIAD

G#	C	D#
Red	Green	Indigo
TONIC	FIFTH	THIRD

60 · 30 · 10 (tonic/fifth/third)

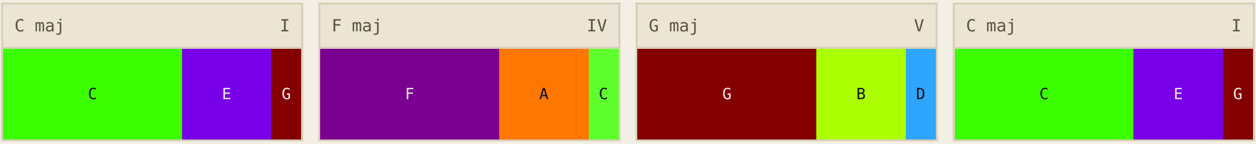
PART VII

Progressions

Harmonic progressions as sequences of these palettes. This is where the second thread — color moving over time, the way music changes key — becomes literal.

I – IV – V – I

C major · The foundational Western loop



The complete loop. *Tonic establishes the world (green, stable, daylit). Subdominant moves away — the camera pulls back, context widens. Dominant introduces urgency, unresolved tension. Return to tonic closes the circle: relief, resolution, homecoming. In a comic panel sequence: establishing shot → widening context → close tension → resolution beat.*

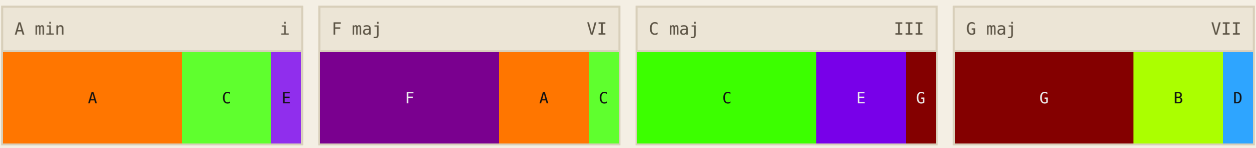
I – V – vi – IV

C major · The ubiquitous pop arc

The emotional loop that never fully resolves. *Tonic → dominant → relative minor → subdominant, then back. The relative minor (A min, vermilion-indigo) introduces shadow without committing to tragedy. Used in hundreds of pop songs for exactly this reason: the sadness is real but the world stays warm. Visually: the same palette cycles, shifting temperature at the third beat — effective for montage sequences and emotional flashback.*

i – VI – III – VII

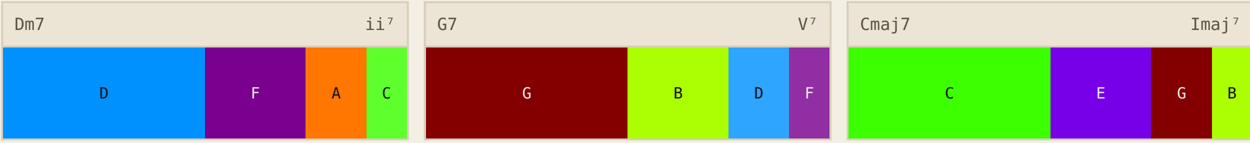
A minor · Cinematic unease



Tension sustained, never released. *The minor tonic stays unresolved across all four beats. The major chords (VI, III, VII) offer false warmth — brief yellow-green and red flares against the cool minor foundation — but the progression circles back without landing. Classic film-noir and thriller palette logic: something warm keeps appearing, but the dominant register is always cool and unresolved.*

ii – V – I

D min → G7 → C maj · Jazz cadence



The jazz resolution — delayed but inevitable. *The supertonic minor (D min, blue-indigo) creates gentle tension. The dominant seventh (G7, red-orange-green-amber) loads the harmonic spring fully, its four notes spanning an unusually wide color range. The major seventh resolution (Cmaj7, green-cyan-blue-violet) arrives as a sophisticated landing, cooler and more complex than a simple triad. Cinematically: the build through an investigation, the climax, the quiet aftermath. The cool colors of Cmaj7 feel earned.*

12-Bar Blues

C · I–I–I–I / IV–IV–I–I / V–IV–I–V · Four visual bars

The blues as color cycle. *The blues form is not about resolution — it is about a particular kind of tension that is the genre's emotional substance. Four bars of tonic (green, occupying 60% of the color space) establish the world. Two bars of subdominant (deep violet) introduce the ache — the furthest cool note from the tonic. Two bars return to tonic. The turnaround (V–IV–I–V) crests in warm red before circling back. In film terms: the blues palette is an establishing environment that returns to itself. The darkness is not climactic; it is the texture of ordinary life.*

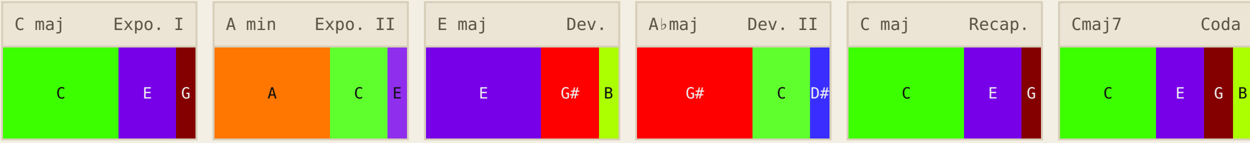
Rhythm Changes — A Section (Bebop)

B♭ · I–vi–ii–V · Fast harmonic rhythm, rapid color cuts

The bebop edit. *Rhythm changes cycle through four chords at high speed — typically two beats each in performance. The color cuts are rapid and the temperature is constantly shifting: warm B♭ major, vermilion-indigo minor, blue-green minor seventh, back to warm dominant. In film terms: this is montage logic — quick cuts that create emotional momentum through juxtaposition rather than through sustained color fields. Each color is too brief to settle; the eye keeps moving. This is how bebop feels: never resting, continuously surprising, the harmonic motion itself is the content.*

Symphonic Arc — Four-Movement Palette

Exposition → Development → Recapitulation → Coda · C major



The long form. *A classical first movement in four structural phases. The exposition establishes tonic (green, stable) and introduces the second subject — here the relative minor, moving into cool blues. Development departs into foreign keys: E major, then A♭ major — the chromatic extremes, violet and deep red, the furthest from home. Recapitulation returns both subjects to the tonic key: the world restored, but the memory of distance still audible in the harmony. The coda resolves fully to the tonic triad. Visually: this arc describes the color logic of a feature film — a world established, pushed into darkness or extremity, then restored with accumulated knowledge.*

PART VIII

Visual Narrative Use

The same progressions read as visual story beats: how a shifting palette can carry a narrative arc the way a chord progression carries a musical one.

Four-Bar Blues — The Recurring Loop

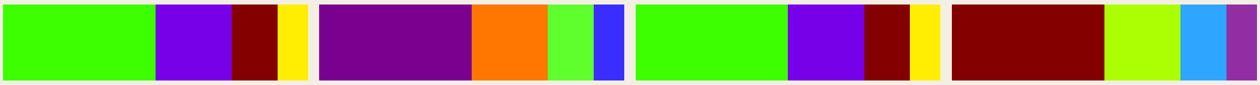
I – I – IV – V · Emotional texture, not resolution

Bar 1–2 Establish

Bar 3 Subdominant

Bar 4 Return

Turnaround V–IV–I



The world. Green dominates — stable, present. The 60% tonic is the scene we live in.

The ache enters. Deep violet is the furthest cool note from the tonic. Something is wrong — but it is ordinary.

Back to tonic. The blues does not resolve; it returns. The world is the same, but you felt it.

Red-orange crests. Then the cycle begins again. The blues is a loop, not an arc.

Visual narrative use: *The blues palette is not for transformation — it is for texture. Use a blues arc when the scene's color should describe an emotional state that does not change, only deepens. The green-violet oscillation is the color equivalent of the riff: familiar, true, and slightly aching.*

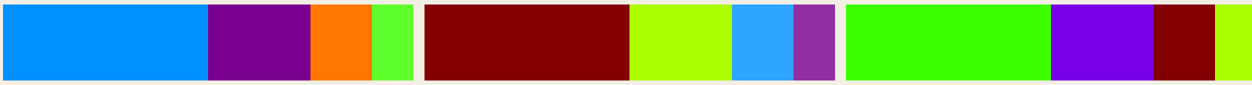
ii – V – I — The Jazz Reveal

Tension building toward a cool, sophisticated resolution

ii min^7 Gather

V^7 Peak tension

Imaj^7 Resolution



Blue-indigo minor. Pieces assembling. Not yet tense — watchful.

Widest color range of the three chords — red through amber. The spring is loaded.

Green-cyan-blue-violet. Cool, complex, resolved. The answer was always in the cool register.

Visual narrative use: *Effective for investigative or intellectual reveals — scenes where the payoff is understanding, not action. The resolution lands in cool colors, which recede visually. The scene relaxes; the viewer exhales. Works for the final panel of a mystery arc.*

Symphonic Development — The Chromatic Journey

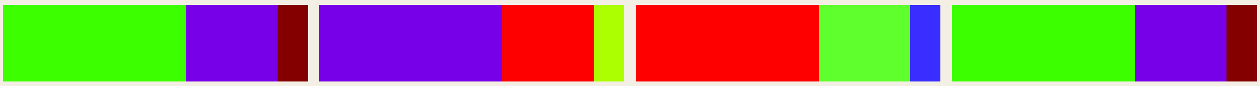
Tonic → Foreign keys → Return · Long-form color arc

Exposition Home

Development I Departure

Development II Darkest point

Recapitulation Return



Green. The world as it is at the story's opening. Stable. The hero in their ordinary life.

Violet. The furthest warm note from the cool tonic. The world has shifted. Something has broken.

Deep red. The chromatic extreme. Crisis, confrontation, the nadir.

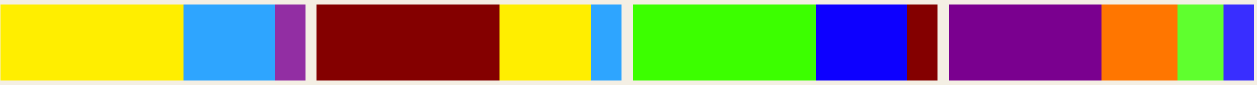
Green again. But the memory of violet and red is in the harmony. The world is restored; the person is changed.

Visual narrative use: *This is the feature film arc. The color palette follows the emotional journey: establishment, departure into extremity, return transformed. A cinematographer using this logic would begin and end in the green register, pushing into warm reds and violets at the story's crisis. The return to green reads as home — but the audience knows what lies behind it.*

Bebop Rhythm Changes — The Rapid Cut

I – vi – ii – V · Montage speed, constant temperature shift

I Warm base | vi Shadow | ii Turn | V^7 Drive



B♭ major: amber-yellow dominant. The warm frame.

G minor: deep red with a cool underside. Brief darkness.

C minor: green-blue. The turn. Color cools rapidly.

F7: violet to deep violet. Drives back to the top. No rest.

Visual narrative use: *Bebop color logic is montage logic. Each beat occupies the frame briefly; the meaning comes from the juxtaposition of temperatures, not from sustained fields. Use this palette strategy for chase sequences, rapid intercutting, or any scene where restlessness itself is the content. The eye cannot settle; neither can the protagonist.*

PART IX

The Harmonic Series

Structure Within the Visible Scale

When a string vibrates at 196 Hz (G3, the anchor), it does not produce a single pure frequency. It simultaneously vibrates in halves, thirds, and fourths — producing a family of overtones called the harmonic series:

HARMONIC	MULTIPLIER	FREQUENCY (FROM G3)	NOTE	CHARACTER
1st (fundamental)	×1	196.0 Hz	G	Strongest
2nd	×2	392.0 Hz	G (octave)	Most consonant
3rd	×3	588.0 Hz	D	Perfect fifth
4th	×4	784.0 Hz	G	Second octave
5th	×5	980.0 Hz	B	Major third
6th	×6	1176.0 Hz	D	Perfect fifth
7th	×7	1372.0 Hz	F	Slightly flat — outside equal temperament
8th	×8	1568.0 Hz	G	Third octave

The harmonic series explains why the interval G-to-D (the third harmonic) is the most fundamental consonance after the octave: D appears in the natural resonance of G. The G major chord (G-B-D) mirrors the first three distinct pitches of G's overtone series, which is why it sounds natural and stable.

Notice what this reveals about the visible scale. The warm-register notes (G, A, A♯, B) include the anchor and its most natural harmonic partners — the notes that ring most strongly when G vibrates. The cool-register notes (C through F) are the notes of the tonic key built on C, a perfect fifth above the anchor. The two registers are not arbitrary halves of the spectrum — they correspond to the two most fundamental tonal centers that emerge from the anchor's harmonic series.

The question for color: do harmonic relationships between notes, translated into light, explain why certain color combinations feel coherent?

The visible scale has two distinct tonal centers — G (the anchor) and C (the tonic) — and the interval structure looks different depending on which one you measure from. Both are shown below.

Intervals from the anchor (G/deep red):

INTERVAL	RATIO	MUSICAL QUALITY	G → NOTE	COLOR	COLOR RELATIONSHIP
Octave	2:1	Perfect consonance	G	Deep Red (696nm)	Monochrome — same color
Major 3rd	5:4	Bright consonance	B	Harlequin (552nm)	Warm → warm register edge
Perfect 4th	4:3	Open consonance	C	Green (521nm)	Warm → tonic (first cool)
Perfect 5th	3:2	Strong consonance	D	Blue (465nm)	Warm → cool register
Tritone	45:32	Maximum dissonance	C♯	Cyan-Blue (492nm)	Warm → cool (visible)

The G major chord (G–B–D) maps to deep red, harlequin, and blue — spanning the anchor, the warm register's edge, and the first strongly cool note. The perfect fourth above G lands at the tonic (C/green), the exact boundary between warm and cool. The perfect fifth crosses into the cool register. Even the tritone above G lands within the visible spectrum — as cyan-blue.

Intervals from the tonic (C/green):

INTERVAL	RATIO	MUSICAL QUALITY	C → NOTE	COLOR	COLOR RELATIONSHIP
Octave	2:1	Perfect consonance	C	Green (521nm)	Monochrome — same color
Minor 3rd	6:5	Soft consonance	D♯	Indigo (439nm)	Cool → cool
Major 3rd	5:4	Bright consonance	E	Violet (414nm)	Cool → deep cool
Perfect 4th	4:3	Open consonance	F	Deep Violet (391nm)	Cool → spectrum edge
Perfect 5th	3:2	Strong consonance	G	Deep Red (696nm)	Cool → warm anchor
Tritone	45:32	Maximum dissonance	F♯	Near-UV (369nm)	Cool → beyond visible

From the tonic, every consonant interval lands within the visible spectrum. The tritone alone — the most dissonant interval in Western music — reaches past the boundary of vision. Every other interval from the tonic lands within the visible scale.

> **A note of caution:** Color harmony as established by color science operates through opponent-process mechanisms in the retina and visual cortex — not through frequency ratio computation. The reason complementary colors create visual tension has been explained by Hering's opponent-color theory (1892) and confirmed by neurophysiology, without reference to frequency ratios. The harmonic series explains acoustic consonance through overtone alignment and acoustic beating — a mechanism that has no direct parallel in how the visual system processes simultaneous wavelengths. The interval-to-color mapping is a structural observation, not a claim about perceptual mechanism.
>
> *Entry of Inquiry:* Whether the opponent-color visual pathway could be responding to frequency ratios between light wavelengths — not as a direct mechanism but as an evolved heuristic — is worth investigating. Both the auditory and visual cortex perform frequency decomposition of their inputs using analogous computational strategies (DeValois & DeValois, 1988). Whether this shared architecture reflects common evolutionary optimization is an open question.

Sources for Part IX: Standard acoustics for the harmonic series; DeValois, R.L. & DeValois, K.K. (1988). Spatial Vision. Oxford University Press.

PART X

The Greek Tetrachord

Ancient Theory and the Cool Register

The visible chromatic scale's cool upper register — C, C♯, D, D♯, E, F — is not a random set of notes. It contains the Greek diatonic tetrachord (C, D, E, F), plus D♯ — the movable inner note that defines the chromatic genus — and C♯, which occupies the inter-genus tonal space between the diatonic inner note (D) and the chromatic inner note (D♯).

The Three Genera

The Greeks did not have one tetrachord — they had three, produced by placing the two movable inner notes at different positions within the fixed outer frame of C and F:

GENUS	INTERVALS	NOTES (APPROX.)	CHARACTER
Diatonic	Whole, whole, semitone	C D E F	The oldest and most natural genus. Aristoxenus called it "of everyday melody." The Babylonian system used the same structure a millennium earlier.
Chromatic	Minor third, semitone, semitone	C D♯ E F	Dense cluster of notes at the bottom with a wide leap to F. Named for the use of altered (*chroma*) tones.
Enharmonic	Major third, quarter-tone, quarter-tone	C C+ D♯♯ F	"Most refined and difficult to hear." Two notes packed into a narrow cluster with a large leap.

All three genera share the same outer frame: C and F — both visible, green and deep violet. D♯ is precisely the note the Greeks moved when constructing the chromatic genus (C, D♯, E, F). C♯ occupies the inter-genus space between the diatonic D and the chromatic D♯ — present in the visible band, though not assigned to any of the three standard genera. The visible spectrum contains the complete tonal material of the diatonic and chromatic genera, plus the tonal space between them that the enharmonic genus approached differently through quarter-tones outside the 12-note system.

This section of the visible scale — from green (C) to deep violet (F) — is bounded at its lower end by the perfect fifth above the anchor (the G-to-C interval), and at its upper end by the note that sits at the threshold of vision (F). It is a structurally coherent region of the chromatic scale, independently identified by the oldest surviving music theory tradition as the foundational unit of musical structure.

Pythagoras and the Music of the Spheres

Pythagoras proposed that the same ratios governing musical consonance — 2:1, 3:2, 4:3 — govern the proportions of the cosmos. He called this *musica universalis*: the music of the spheres. His core insight was that these ratios are not human inventions but discovered properties of reality.

The octave-doubling method this investigation uses is, in one sense, a literal application of the Pythagorean intuition: that the octave relationship is universal, connecting different scales and different domains of reality. Following it approximately 41 octaves upward from G3, we arrive in visible light — with G3 at red and the tetrachord in the cool upper register.

Whether Pythagoras was right that this extension is physically meaningful, or whether the mathematical elegance of the result is a property of the mathematics rather than of nature, remains the central open question.

Sources for Part X: Aristoxenus, Elementa Harmonica; standard classical music theory scholarship.

PART XI

Independent Convergence

Cultures That Found the Same Notes

The visible chromatic scale contains notes that are acoustically fundamental across the full spectrum: the warm register holds G (the dominant — the most important note after the tonic), A (the universal reference pitch), and B (the leading tone); the cool upper register holds C, D, E, and F — the diatonic tetrachord. If these notes are physically privileged across both domains — visible light and acoustic consonance — then independent musical cultures, working without knowledge of each other, might be expected to converge on them independently. The evidence is striking.

Ancient Mesopotamia (c. 2000–1400 BC) Cuneiform tablets from Ugarit (c. 1400 BC) contain the oldest surviving notated melody, built on a heptatonic scale described in terms of overlapping tetrachords. The Babylonian system predates Greece by over a millennium and was already diatonic — using the same whole-tone, whole-tone, semitone structure as the Greek diatonic tetrachord. This parallel was established by 20th-century musicology, notably by Anne Draffkorn Kilmer and M.L. West, after the cuneiform tablets were deciphered. Aristoxenus, writing in the 4th century BC, had no access to Babylonian sources.

Ancient China: The Gong System (c. 500 BC) The Chinese pentatonic scale (Gong, Shang, Jue, Zhi, Yu — roughly C, D, E, G, A) includes three of the four diatonic tetrachord notes: C, D, and E, all in the cool visible register. Confucius mapped these notes to social hierarchy. The five-element system explicitly mapped colors, notes, seasons, and organs to the same five categories — a cross-domain mapping that this investigation approaches from a different direction.

Native American Plains Peoples Tetratonic (four-note) scales were common among the Arapaho, Blackfoot, Crow, Omaha, Kiowa, Pawnee, and Sioux. Ethnomusicologists note these scales frequently span a perfect fourth — precisely the C-to-F range of the diatonic tetrachord, the cool register’s outer frame. Independently developed on a separate continent, the same four-note, perfect-fourth structure recurs.

Maori (Aotearoa New Zealand) A 1969 ethnomusicological study by Mervyn McLean analysed 651 Maori scales and found tritonic (three-note) scales most common (47%), followed by tetratonic (31%), and ditonic (two-note) scales at 17% (McLean, 1969; McLean, 1996). The tritonic scales most commonly used major second and minor third intervals — producing C, D, and approximately D♯/E♭ — notes of the cool register.

Gregorian Chant (c. 600–1200 AD) Built on a system of eight modes, each defined by its relationship to a tetrachord. The theoretical framework explicitly constructed scales from stacked tetrachords. The church modes are direct descendants of Greek modal theory, inheriting the tetrachord as their fundamental structural unit.

A note of caution: The global prevalence of tetratonic and pentatonic scales is well-documented but has a compelling explanation that requires no appeal to light frequencies. The harmonic series, vocal physiology, and the natural resonances of simple instruments all independently favor small pitch sets organized around simple frequency ratios. The tetrachord emerges from acoustic physics alone. The coincidence that the acoustically privileged notes and the cool-register visible notes overlap may simply reflect that both are constrained by the same harmonic series — not that there is any direct relationship between sound and light. This objection is well-founded and likely at least partially correct. The most honest account is probably that the harmonic series produces both the acoustically privileged notes and the visible-register notes through different pathways, making their convergence mathematically expected rather than physically mysterious.

Sources for Part XI: Everett, D.L. (2005). "Cultural Constraints on Grammar and Cognition in Pirahã." Current Anthropology, 46(4), 621–646. McLean, M. (1969). "An analysis of 651 Maori scales." Yearbook of the International Folk Music Council, 1, 123–164. DOI: 10.2307/767637. McLean, M. (1996). Māori Music. Auckland University Press. ISBN: 978-1-86940-144-3. Hurrian Hymn No. 6 (H6), c. 1400 BC, for Babylonian precedent. Kilmer, A.D. (1974). "The Cult Song with Music from Ancient Ugarit." Revue d'Assyriologie, 68, 69–82. West, M.L. (1992). Ancient Greek Music. Oxford University Press. Aristoxenus, Elementa Harmonica.

PART XII

Prior Musical Color Theories

How this mapping sits against its predecessors — Newton, Castel, Rimington, Scriabin, Helmholtz, and Cousto's Cosmic Octave — and what, precisely, is and isn't new here.

1704

Newton's Prism Scale

Isaac Newton — Opticks

Newton divided the prismatic spectrum into seven named colors to match the seven notes of the diatonic scale — adding Orange and Indigo specifically to reach the required count. The analogy was explicit: violet recurring at the far end of the spectrum corresponded, he felt, to the octave recurrence in music. The physics of the prism was used as post-hoc support; the mapping direction was musical → color, not the reverse.

METHOD	Aesthetic/analogical — 7 colors forced to match 7 diatonic notes
ANCHOR	None fixed — proportional band widths matching musical intervals
CHROMATIC?	No — diatonic only (7 notes)
PHYSICS-DERIVED?	No — imposition, not derivation
KEY PROBLEM	Indigo invented to make the numbers work

1856–1867

Helmholtz Color-Music Scale

Hermann von Helmholtz — Handbook of Physiological Optics

Helmholtz anchored G to the Fraunhofer A line — a dark absorption band in deep red — and ran the diatonic white-note scale upward from there: G = deep red, A = red, B = orange, C = yellow, D = greenish-blue, E = indigo-blue, F = violet, g = ultra-violet. He used physical reasoning about wavelength correspondence, not aesthetic choice. Notably, his full chromatic extension continues past the octave — g♯, a, a♯ all fall in ultra-violet — meaning his scale extends further than this work's 11-note result. The specific wavelength of his Fraunhofer A-line anchor has not been confirmed identical to G3 = 696nm, but the direction and physical reasoning converge. Helmholtz also noted that complementary color pairs — such as red and blue-green — were analogous to consonant musical intervals.

METHOD	Wavelength correspondence — frequencies octave-transposed to light range
ANCHOR	G → Fraunhofer A line (deep red) — same direction as G3 = 696nm; exact wavelength unconfirmed
CHROMATIC?	Partial — white notes only (7 diatonic), chromatic extension goes past the octave into UV
PHYSICS-DERIVED?	Yes — physical reasoning; converges on same starting direction, different method
KEY LIMIT	Scale extends beyond the octave; does not isolate the 11-note visible chromatic result

c. 1910

Scriabin's Color-Key System

Alexander Scriabin — Prometheus: Poem of Fire

Scriabin mapped all twelve keys to colors around the circle of fifths, based on his synesthetic experience. The result is a complete chromatic system, but the assignments are personal and esoteric — C major is reported as red, G major as orange-rose, D major as yellow, and so on around the cycle. (Scriabin's exact assignments are documented inconsistently across sources, and his system evolved over his lifetime; the specific colors should be treated as approximate.) Keys a fifth apart receive similar colors; keys a tritone apart receive opposing colors. There is no physical principle governing the specific assignments, though the circle-of-fifths structure imposes a kind of harmonic logic on the color sequence.

METHOD	Synesthetic perception + circle of fifths organization
ANCHOR	Personal — C major reportedly red; assignments vary across sources and evolved over time
CHROMATIC?	Yes — all 12 keys mapped
PHYSICS-DERIVED?	No — phenomenological, not physical
KEY LIMIT	Maps keys, not individual notes; non-reproducible

1893 / 1912

Rimington's Colour Organ

Alexander Wallace Rimington — Colour-music: The Art of Mobile Colour

Rimington built a keyboard instrument — the clavier à lumières — that projected colored light in correspondence with musical keys. He mapped octaves of the color spectrum to octaves of the musical scale and argued that color harmony follows the same laws as musical harmony. His system was performative rather than analytical, and the specific note-color assignments were chosen for perceptual effect rather than derived from first principles.

METHOD	Performative — keyboard controlling projected colored lights
ANCHOR	Octave-color analogy, pragmatically assigned
CHROMATIC?	Partial — focused on keys and octave registers
PHYSICS-DERIVED?	No — aesthetic and performative reasoning
KEY LIMIT	No reproducible physical grounding; system never standardized

1978–1984

The Cosmic Octave

Hans Cousto — Die Kosmische Oktave / The Cosmic Octave: Origin of Harmony

Cousto, a Swiss mathematician, formulated the general method this work uses years before this project existed: take any periodic phenomenon — a planetary orbit, a color's frequency, a molecular vibration — and repeatedly halve or double it by octaves until it lands in the audible or visible range. He self-published the method specifically applied to sound and color in a 1980 bilingual booklet, then expanded it into a full book in 1984. This is the closest and oldest precedent for the core mechanism (octave-transform light frequency to compare it against sound), predating this project by roughly 48 years. Cousto's project is broader in scope (any periodic phenomenon, not just the chromatic scale) and shallower in music-theoretic detail — it does not derive a 12-note chromatic mapping, does not identify a red-boundary anchor, and does not analyze which specific notes fall inside or outside the visible band. His wider body of work is also associated with "planetary frequency" sound-healing claims that carry no evidentiary support and should not be conflated with the narrower, checkable arithmetic claim.

METHOD	Octave-doubling/halving — identical core operation to this project's $\times 2^n$ transformation, applied to arbitrary periodic phenomena
ANCHOR	None fixed — any starting frequency is octave-transposed into the target range
CHROMATIC?	No — does not derive or analyze a 12-note chromatic mapping
PHYSICS-DERIVED?	Yes — the octave arithmetic is the same core method this project uses; predates it by ~48 years
KEY LIMIT	No red-boundary anchor, no tritone-at-threshold finding, no warm/cool analysis; broader scope, less musical rigor; entangled with unrelated sound-healing marketing claims

2026

Modulations

Modulations — An Investigation into Color, Music, and Time

Starting from a single physical principle — octave doubling — and a single physical fact — the red boundary of the visible spectrum at ~700nm — the full chromatic scale is derived without any aesthetic choices. G3 × 2^{41} = 696nm. Every other note follows from equal temperament arithmetic. The result: 11 of 12 chromatic notes land in the visible spectrum, the diatonic tetrachord (C D E F) sits entirely in the cool register, and the warm/cool boundary falls at the perfect fifth (G→C) — the most fundamental interval in Western harmony after the octave.

METHOD	Octave-frequency arithmetic — $\times 2^{41}$ applied to all 12 chromatic notes
ANCHOR	G3 = red boundary of visible light (431 THz / 696nm) — physically determined
CHROMATIC?	Yes — 11/12 notes fully in spectrum, 12th at UV threshold
PHYSICS-DERIVED?	Yes — no aesthetic choices made
KEY RESULT	Warm/cool boundary = perfect fifth; tetrachord in cool register; robust across 11 tuning standards

HELMHOLTZ CONVERGENCE — WHAT IT DOES AND DOES NOT SHOW

Helmholtz, working from physical reasoning in the 1860s, independently anchored G to the deep red end of the spectrum — the same direction this work arrives at from octave-frequency arithmetic. This is the most significant finding in the prior-theory survey: the greatest acoustician of the 19th century used a physical principle (wavelength correspondence) rather than an aesthetic one, and landed in the same spectral region. The precise wavelength of his Fraunhofer A-line anchor has not been confirmed identical to G3 = 696nm, and his scale extends past the octave into ultra-violet rather than isolating 11 visible notes. The convergence is of method and direction — not of identical result. He did not arrive at the full chromatic scale, the 11-note finding, or the warm/cool boundary. Those are the contributions of this work.

THE COUSTO PRECEDENT — WHAT IT DOES AND DOES NOT SHOW

Cousto's "law of the octave" (1978) is the closest existing precedent for this project's core mechanism, and it predates this project by roughly 48 years. It shows that treating light as many octaves above audible sound, and using repeated halving/doubling to bring one into comparison range with the other, is not a novel move — it has an established history, including a patented implementation (see Sources) and independent rediscoveries by physicists and hobbyists in the decades since. What Cousto's system does not do is apply the method to the full 12-note chromatic scale, identify a physically-motivated anchor at the red boundary, or derive the tritone-at-threshold finding. This project's contribution is not the octave-transposition method itself — that credit belongs to Cousto and, in narrower form, to Helmholtz — but the specific, checkable, musically detailed result obtained by applying that method rigorously to equal temperament. This distinction should be stated plainly in any external presentation of the project: the mechanism is not new; the chromatic-scale result obtained from it is what this project adds.

PART XIII

Non-Musical Art Theory

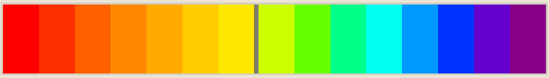
Where the derived mapping meets established color theory: the warm/cool structure, its boundary at the perfect fifth, and where the two frameworks agree and diverge.

The theories in Part XII all begin with music and ask where the colors land. A different and more revealing question is whether the major traditions of non-musical color theory — those developed by artists and educators entirely independently of acoustics — arrive at any of the same structural conclusions. The answer is specific: one convergence is real and worth stating carefully. The rest would be forced.

The genuine convergence is the **warm/cool boundary**. Art theory arrived at it by intuition and perceptual psychology; this work arrived at it by acoustic physics. They land in the same place.

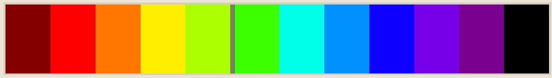
THE ITTEN / HAYTER WARM–COOL DIVIDE

Art Theory — Itten (1961), Hayter (1813)



Charles Hayter first formalized a warm-cool division in a hue circle in 1813, placing the dividing line at **yellow-green / purple-red**. Johannes Itten — the Bauhaus colorist who became the most influential voice in 20th-century color education — adopted the same boundary in 1961. The warm side: reds, oranges, yellows. The cool side: greens, blues, purples. The boundary was drawn by perceptual psychology — what feels warm versus cool to the eye — not by any physical measurement.

Modulations — G3 anchor, 2026



The physics-derived map places the warm register at G through B (deep red → harlequin) and the cool register at C through F (green → deep violet). The boundary falls at **the perfect fifth (G→C)** — the most fundamental interval in Western harmony after the octave. To be precise: the spectral boundary sits between adjacent notes B (552nm) and C (521nm), a semitone gap of ~31nm. But the harmonic significance of that boundary is the G→C relationship underlying it — the anchor-to-tonic interval. The semitone is where the boundary sits in the spectrum; the perfect fifth is what it means. This boundary was not chosen; it emerged from the arithmetic. In spectral terms it lands at approximately 520–555nm: the yellow-green / green transition. This is the same spectral location Hayter and Itten placed their boundary by intuition, two centuries apart.

WHAT THIS CONVERGENCE DOES AND DOES NOT SHOW

The warm/cool boundary is where acoustic physics and artistic intuition touch the same point on the spectrum. Both systems agree that the spectrum divides into a red/orange/yellow group and a green/blue/violet group, with the boundary somewhere in the yellow-green region. This work places that boundary at a musically significant interval — the perfect fifth. Itten placed it at a perceptually significant transition — the point where colors shift from advancing to receding in the visual field. That two independent systems agree on the same spectral location is notable. It does not prove a connection; it invites an explanation.

GOETHE'S LIGHT–DARK POLARITY (PARALLEL, NOT CONVERGENCE)

Goethe organized color in his 1810 *Theory of Colors* around a fundamental polarity of light and darkness — all colors arise from the interaction of these two poles, with yellow on the light side and blue on the dark side. This is structurally parallel to the extended color map's octave-lightness axis, where higher registers (brighter octaves) render in lighter tints and lower registers in darker shades. Both systems use light/dark as the second organizing dimension of color, orthogonal to hue.

This is worth noting as a structural parallel, not as a convergence. Goethe's polarity is about emotional and perceptual qualities of color; the octave-lightness axis is about encoding acoustic register. The structural resemblance suggests both systems were responding to the same underlying perceptual reality — that brightness and darkness carry information independent of hue — rather than that one derived from the other.

Note: No non-musical color theorist independently arrived at groupings equivalent to a major triad, a chord progression, or a harmonic function. The chord structures produced by octave-frequency mapping are genuinely novel to this mapping and have no parallel in visual art theory. That absence is honest and worth preserving.

PART XIV

How the Mind Carves a Continuum

Color and Music as Parallel Acts of Perception

The World Color Survey — a systematic study of color naming across 110 unwritten languages — found a universal sequence in how cultures develop color vocabulary: black/white first, then red, then yellow or green, then the other, then blue, then brown, then the remaining colors. Blue is always last among basic color terms. Homer's Iliad never uses a word for blue despite describing the sea extensively — though ancient Greek had blue-adjacent terms (kyanos, glaukos), it lacked a basic color term for blue in the technical sense (Berlin & Kay, 1969; Kay et al., 2009; Gladstone, 1858).

A parallel progression appears in musical pitch systems worldwide:

STAGE	COLOR VOCABULARY	MUSIC PITCH SYSTEM
1	Light/dark only	Rhythm only — no pitched notes
2	+ Red	Ditonic — 2 pitches
3	+ Yellow or Green	Tritonic — 3 pitches
4	+ the other of Y/G	Tetratonic — 4 pitches (tetrachord range)
5	+ Blue	Pentatonic — 5 pitches (most widespread)
6	+ Brown	Heptatonic — 7 pitches (diatonic)
7	Full vocabulary	Chromatic — 12 pitches (full Western system)

Specific cultural cases fit the pattern: the Pirahã have among the fewest basic color terms documented and highly constrained vocal music with no fixed pitch system (Everett, 2005). The Himba have elaborate green terms but no blue term; their music system has not been sufficiently documented to make a specific pairing claim here. Ancient China had five color terms and a pentatonic scale. Ancient Greece had four reliable color terms and a tetrachordal foundation. Western culture has the most elaborate color vocabulary and a twelve-tone chromatic system.

The direction of this parallel is what the frequency framework predicts. Warm colors first, cool colors last — which aligns with warm notes (G through B, the lower visible register) being the simpler, more immediately physical anchor, and cool notes (C through F, the upper visible register, the tetrachord) being the more refined structure named later in cultural development.

Beneath the Parallel: A Single Cognitive Operation

Stated only as a parallel, this is a little superficial. Two systems that both grow from simple to complex is not, by itself, remarkable. If the correspondence between color and music stopped at "both get more complex," it would be a curiosity, not a finding.

But there is a deeper layer, and it is the most solid cross-domain result in this entire investigation. It does not depend on the octave-doubling arithmetic at all. It is this: **color and music are both continuous physical dimensions that the human mind divides into a small number of discrete categories — and it divides them the same way, by the same cognitive operation, with the same two-layer structure of universal anchors and culturally-drawn boundaries.**

This operation has a name in cognitive science: categorical perception.

The Rainbow Is Not Continuous to You

Physically, the visible spectrum is a smooth gradient. The wavelength changes continuously from 700nm to 380nm with no breaks, no bands, no lines. And yet no one sees it that way. Everyone sees a rainbow — discrete stripes of red, orange, yellow, green, blue, violet. The bands are not in the light. They are imposed by the perceptual system.

This is categorical perception, and its signature is precise and measurable: the wavelength difference between a blue and a green looks much larger than an equal-sized wavelength difference between two shades of blue within the blue band. Physically equal steps are perceived as unequal — compressed within a category, expanded across a boundary.

Music Does the Same Thing

Pitch is also a continuous dimension. Yet no musical culture uses the continuum — every culture selects a small number of discrete pitches. The strongest evidence that this is the same cognitive operation as color categorization comes from a remarkable cross-cultural experiment. Researchers used an iterated reproduction method — a kind of "telephone" game — across 39 participant groups in 15 countries, spanning urban societies and Indigenous populations. Across every single group tested, random continuous input collapsed toward discrete categories sitting at small-integer ratios. The specific categories varied across cultures, often matching local musical practice.

That is exactly the structure of color categorization: a universal tendency to carve the continuum, anchored at perceptually privileged points, with the specific boundaries filled in by culture.

The Two-Layer Structure

For decades, color category research was framed as a war between universalists (Berlin & Kay: categories are biologically fixed) and relativists (Berinmo, Himba: categories are culturally constructed). The resolution, articulated by Regier and Kay in a 2009 review titled "Whorf was half right," is that both camps were partly correct. Universal perceptual anchors exist — the focal colors sit where visual neurophysiology predicts — but the boundaries drawn between categories vary by culture.

Universal skeleton. Cultural flesh.

The cross-cultural rhythm study found exactly the same structure: a universal tendency toward discrete small-integer-ratio categories (the skeleton), with the specific categories varying by local musical tradition (the flesh). This is the genuine unifying principle: **the human mind processes both color and pitch through the same cognitive architecture — anchored by biology, bounded by culture — applied to two different perceptual continua.**

A Note on Color Naming and Musical Standardization

One asymmetry deserves explicit acknowledgment. Musical note frequencies are standardised by international agreement: A4 = 440 Hz (ISO 16:1975). From that single anchor, all twelve notes of the equal-tempered octave are defined to exact frequencies. When a piano in Tokyo and a piano in Berlin play "A," they produce the same frequency.

No equivalent exists for color. No international body has decreed that "blue" means a specific wavelength. The word "blue" covers a broad, culturally-defined band whose edges are matters of language, not physics. This is why two coloring pencils both labeled "blue" can differ visibly: no standard was violated, because no standard exists.

The visible light spectrum is physically just as precise as the audible frequency range — 470nm light is as well-defined as 440 Hz sound. The imprecision is entirely in the naming. This investigation, by deriving color assignments from note frequencies, effectively lends music's standardization to color: each note gives a single, exact, reproducible wavelength target, more precise than any color name has ever carried.

> **A note of caution:** The two-layer structure is the mainstream consensus in color science, but the relativist evidence (Berinmo, Himba) is real and should not be minimized. The music side is less established than the color side — the claim that pitch-scale formation is literally the same operation as color categorization is an inference from parallel structure, not a directly demonstrated identity.
>
> *Entry of Inquiry:* If color and pitch are partitioned by the same category-forming operation, the same individuals who draw sharper color-category boundaries should draw sharper pitch-category boundaries. Does training in one domain transfer to the other? This is testable and has not been done.

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PART XV

The Association Layer

Color, Music, and the Biology of Loaded Meaning

There is a claim you have probably encountered. It states that McDonald’s uses red and yellow because those colors make people hungry. It is repeated in marketing textbooks, design blogs, and casual conversation with the confidence of an established fact. It deserves to be examined — because the examination turns out to be more interesting than the debunking.

Trace the hunger claim to its source and it does not lead to a controlled experiment. It leads to marketing copy. The specific assertion that yellow stimulates appetite rests on no replicable physiological mechanism. Research findings on the appetite-enhancing effect of red and the appetite-reducing effect of blue have been inconsistent across studies. Where color does influence food behavior, the strongest candidate mechanism is not wavelength physics — it is conditioning. Seeing the McDonald’s logo makes people who grew up with McDonald’s think of McDonald’s food. The yellow is doing Pavlovian work. It is not reaching directly into the hypothalamus and switching on hunger.

This matters because it reveals what color-emotion associations actually are. A major cross-cultural study tested color-emotion links across 4,598 people in 30 nations speaking 22 native languages. It found robust universal associations — but also that similarity was greater when nations were linguistically or geographically close, and that nation predicted associations above and beyond the universal patterns. The structure is universal skeleton, cultural flesh — exactly the same two-layer architecture documented in Part VIII-B for color categories and pitch scales.

The cultural specificity is worth being precise about. The same study found that Germans link envy to yellow far more than any other tested nation. “Yellow means envy” is not a biological universal. It is a cultural artifact specific to a language group. “Yellow means hunger” is almost certainly in the same category — a local association laundered through the language of neuroscience.

What the music evidence actually shows

The claim that there is little intrinsic reason why notes played a certain way should make us feel anything turns out to be only partially correct — and the way it is wrong is more interesting than the original claim.

In 2009, researchers played computer-generated Western piano music to members of the Mafa tribe of Cameroon — a population with no access to Western media, no prior exposure to Western music. The Mafa recognized happy, sad, and fearful expressions in that music above chance. The emotional signal was not culturally acquired. It was read off acoustic features that those listeners had never encountered in that musical tradition.

A 2019 *Science* study by Mehr and colleagues, analyzing ethnographic records and field recordings from societies across the world, found something more fundamental: music appears in every society observed; acoustic features of songs reliably predict their primary behavioral context; and tonality is widespread, perhaps universal. Lullabies are soft and slow across cultures. Dance music is fast and rhythmic. Not because some cultural treaty was signed, but because those acoustic profiles map onto human physiological states that appear to be shared across the species. A commentary on the study put it plainly: human musicality fundamentally rests on a small number of fixed pillars — hard-coded predispositions afforded by the ancient physiological infrastructure of shared biology.

A 2024 *PNAS* study measured bodily sensations evoked by music in 1,938 Western and East Asian participants. Music-induced bodily sensation maps were highly consistent across the two cultures ($r = 0.80–0.91$). Happy music energizes limbs. Sad music settles in the chest. Across both cultures, with both Western and East Asian music.

There are limits to this universality, and they are honest ones. The Fritz 2009 researchers themselves cautioned against conjuring the idea of music as a universal language of emotion, calling this partly a legacy of Romanticism. And work with the Tsimane people of Bolivia found that even octave equivalence — the perception that A4 and A5 are “the same note” — does not appear to hold universally, suggesting some of what Western music theory treats as fundamental may require cultural acquisition. The universality is real and documented, but it operates at the level of basic acoustic-emotional anchors, not at the level of the specific emotional content of any musical tradition.

So for music, as for color: universal skeleton, culturally shaped flesh.

Why culture is not nothing

This is where early developmental research illuminates something that neither the color nor the music literature fully explains on its own.

Dutch adults who had been adopted from Korea as infants — some as young as three to five months old, before any conscious memory is possible — learned to identify an unfamiliar three-way Korean consonant distinction significantly faster than Dutch controls with no Korean exposure. Even those adopted in the first half-year of life showed the advantage. This is not genetics. These adoptees had no continuing exposure to Korean, no Korean-speaking family, no reason to retain anything. What they retained was a neurological trace — a pattern encoded during a sensitive developmental window before language, before memory, before culture in any conscious sense. The finding indicates that phonological knowledge is in place before six months of age.

The biological skeleton is not only present at birth. It is being written in the first months of life, from acoustic input that arrives before the child can do anything about it. The flesh that culture adds later is applied to a substrate that was already partly primed.

This reframes the McDonald's story exactly. The colors are not making people hungry through any biological mechanism. They are triggering a conditioned response built on top of a genuine but modest biological sensitivity — the arousal and attention-capture effects of saturated, high-contrast color — that was then loaded, through decades of repetition, with the specific content of fast food. McDonald's didn't choose yellow because yellow causes hunger. They chose a high-visibility combination, and then caused an association through sheer repetition. The biology enabled the loading. The culture did the loading. These are not the same thing, and conflating them is what makes the marketing claim sound more scientific than it is.

What this layer adds to the frequency framework

The frequency framework in this investigation operates at the level of the skeleton: it asks what frequencies are available, where the auditory and visual ranges sit relative to each other, what mathematical structure organizes those ranges. The cultural loading — what specific meaning gets attached to which frequency — is downstream of that. The architecture comes first. The meaning follows.

Both color and music have biological anchors that set the range of what is available for cultural meaning. Those anchors are real, cross-cultural, and modest. Culture fills the available space with specific content through early exposure, repetition, and reinforcement. Once loaded, the response is involuntary. The person who tears up at a minor key, or who instinctively associates a color with a memory, is not choosing that response. They are experiencing an imprint that was installed before they could interrogate it.

> **A note of caution:** The claim that color-emotion associations and musical emotional responses share the same two-layer structure is an inference from parallel evidence, not a direct demonstration. Each domain has its own literature; the cross-domain synthesis is this project's interpretation. The individual findings are well-established; the synthesis belongs to the project.

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PART XVI

Mathematical Predictions

Colors and Notes Not Yet Named

If the parallel between color vocabulary and musical complexity is real, the harmonic series can make specific predictions about frequencies waiting to be culturally recognized.

The 11th and 13th harmonics produce pitches that exist between the notes of the Western chromatic scale — genuinely new frequency locations that the harmonic series generates but Western music has not fully incorporated.

To compare these harmonics with the visible-range notes, we reduce them to the same pitch class by dividing by octaves (2^n) until they fall within the audible range of the visible-light mapping. The 11th harmonic from C4 (2877.9 Hz) reduced by 3 octaves = 359.7 Hz; the 13th harmonic from C4 (3401.2 Hz) reduced by 3 octaves = 425.1 Hz. These pitch classes, multiplied by 2^{41}, yield:

HARMONIC	PITCH CLASS (OCTAVE-REDUCED)	HZ	$\times 2^{41}$	WAVELENGTH	IN WESTERN MUSIC?
11th	Between F and F♯	359.7	791 THz	379 nm	No — new note
13th	Between G♯ and A	425.1	935 THz	321 nm	No — new note

> **Note on method:** Octave reduction is musically principled — the harmonic series generates pitch *classes*, and the same pitch class sounds equivalent across octaves. Reducing by octaves preserves the harmonic identity while bringing the frequency into a comparable range. The resulting wavelengths are where these pitch classes would fall in the visible-light mapping, not where the raw harmonic frequencies fall (which land deep in ultraviolet at 47nm and 40nm respectively).

The 11th harmonic pitch class is particularly interesting: at 379nm, it sits just 1nm inside the visible boundary — between F (391nm, deep violet, the last clearly visible note) and F♯ (369nm, the UV threshold). This places a harmonically significant pitch right at the edge of visibility, alongside the tritone. There may be a cluster of acoustically significant notes that all orbit the threshold of vision.

Cyan: The Next Basic Color Term

More immediately testable: the gap between note C (521nm, green) and note C♯ (492nm, cyan-blue) spans a region — roughly 490–510nm — that human eyes can distinguish extremely finely but that color vocabulary has not yet categorized as a basic term in most languages. This is the blue-green region.

The World Color Survey pattern predicts a cyan-type basic color term will be the next to develop in English — digital screen technology, where cyan is already a primary component, is accelerating this. The frequency framework adds a specific note-equivalent: the pitch between C and C♯ is the musical analog of the unnamed cyan region — a frequency that exists but has no standard name in the scale.

Both are the same phenomenon in different domains: a perceivable frequency that culture has not yet fully named.

Sources for Part XVI: Standard harmonic series arithmetic; World Color Survey for cyan prediction.

PART XVII

A Necessary Qualification

This Framework Is Atmospheric

The most important qualification to place on everything above: the framework is not universal. It is specific to Earth's atmosphere, our star, and the evolutionary history of human sensory systems.

Why the Visual Range Is Where It Is

The visible spectrum is not an arbitrary band of electromagnetic radiation. It corresponds to the atmospheric optical window — the narrow spectral region that Earth's atmosphere actually transmits to the surface with high efficiency. Most infrared is absorbed by water vapor and carbon dioxide before reaching the ground. Most UV below approximately 300nm is blocked by ozone. What passes through is visible light, a narrow band determined by the specific composition of Earth's atmosphere.

This means the ~41-octave gap between human hearing and human vision is not a gap between "where light is" (light is everywhere) but between two specific, physically constrained evolutionary targets. Both the auditory range and the visual range were shaped by the conditions of life on Earth's surface — the first by the acoustic environment of terrestrial mammals, the second by the specific spectral window that Earth's atmosphere opens.

The Solar Entropy Connection

Eyes did not evolve to see visible light merely because it was available. They evolved to detect light where the solar information environment is richest. Research by Delgado-Bonal & Martín-Torres (2016) showed that human photopic vision peaks at precisely 555nm — the wavelength at which solar radiation, filtered through Earth's atmosphere, delivers the optimal balance of energy and information content. Scotopic (low-light) vision peaks at 508nm, also derived from the solar information optimum under different atmospheric filtering conditions.

Where does Middle C land in relation to these reference points? Under modern orchestral tuning (A440–A444), Middle C × 2^{41} lands at 517–521nm — within 0–3nm of the unfiltered solar entropy peak (518nm, Delgado-Bonal 2016). Under Baroque historically-informed performance pitch (A415), the same transformation lands at 553nm — within 2nm of the atmospherically-filtered photopic optimum (555nm). The full practical range of Western tuning history, from A415 to A444, maps Middle C to a band of 517–553nm.

This band is precisely the solar-relevant spectral region: the zone where solar entropy production peaks, where human scotopic vision is optimised (508nm), where the unfiltered entropy maximum sits (518nm), and where photopic vision peaks (555nm). It is also the region where a third independent biological system converges: all major plant and phytoplankton pigments — chlorophyll a, chlorophyll b, carotenes, xanthins — have peak absorption between 430 and 550nm, precisely where entropy production due to photon dissipation is maximal for our solar spectrum (Michaelian, 2022). Three independent biological systems converge on this spectral region. Middle C, across Western tuning history, lands within it.

The honest statement is: no single tuning places Middle C precisely on any single solar reference point. But the correspondence is with a physically meaningful spectral region, not a coincidentally chosen wavelength.

> **A note of caution:** This correspondence is structural, not causal. We have described a convergence. We have not explained it. Why the human auditory range should be separated from the visual range by approximately 41 octaves — whether this is derivable from physics, reflects a shared evolutionary optimization, or is an arithmetic coincidence of two independently constrained systems — remains genuinely unknown. This is the project's central unresolved problem.
>
> *Entry of Inquiry:* If the visual range is determined by the solar information environment, and the auditory range was shaped by terrestrial acoustics — could the 41-octave gap be derivable from first principles? No one has attempted to extend the Delgado-Bonal information-theoretic framework to derive an optimal acoustic range from solar physics. If such a derivation independently produced ~20–20,000 Hz, that would constitute a mechanism. Until it exists, the gap remains a description rather than an explanation.

Sound Behaves Differently in Different Media

Sound is a mechanical pressure wave. It requires a medium. On Mars, whose atmosphere is roughly 1% as dense as Earth's, sound barely propagates beyond a few meters — confirmed by NASA's Perseverance rover in 2021. Light requires no medium and crosses vacuum effortlessly. This is the fundamental asymmetry between the two domains.

A creature evolving under a cooler red-dwarf star would have a visual system shifted toward longer wavelengths. The octave-doubling calculation for such a creature would produce different visible notes. The framework is local, not universal — which makes it, if anything, more interesting as a description of the specific sensory world that human consciousness inhabits.

> *Entry of Inquiry:* Earth's atmosphere is changing due to human activity — ozone depletion, aerosol pollution, global dimming and partial recovery since the 1980s (Wild, 2009). The Delgado-Bonal information-theoretic optimum (555nm) was derived from current atmospheric composition. Evolutionary adaptation operates over 100,000-year timescales, not decades, so the immediate concern is UV-related eye damage rather than photoreceptor tuning. But the question of whether the spectral optimum has shifted measurably is honest and worth quantifying.

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Where This Leaves Us: An Honest Summary

Following the mathematics of frequency from audible sound into visible light produces a set of results worth reporting honestly.

What the arithmetic shows, without dispute

G3 (196 Hz) × 2^{41} = 696nm — the deep red boundary of the visible spectrum. From G3, eleven consecutive chromatic notes step through the full visible band: deep red, red, vermilion, amber-yellow, harlequin, green, cyan-blue, blue, indigo, violet, deep violet. The twelfth note — F♯, the tritone — lands 11.3nm past the violet boundary. The visible spectrum is an almost-complete chromatic scale, missing only the most dissonant interval. The warm colors (G through B) occupy the lower register; the cool colors (C through F) occupy the upper. The boundary between them is the perfect fifth. These are arithmetic facts.

What the structure reveals

The warm/cool divide in color corresponds to the lower/upper register of the visible chromatic scale, with the boundary at the most fundamental consonance after the octave. The Greek diatonic tetrachord (C, D, E, F) occupies the cool upper register — bounded below by the perfect fifth above the anchor, bounded above by the sole note that exceeds the visible spectrum. The tritone alone sits past the threshold of vision.

Middle C, across the full range of Western tuning history, lands within the solar-relevant spectral region — the zone where three independent biological systems (human photopic vision, human scotopic vision, photosynthetic pigments) converge. The correspondence is with a region, not a single wavelength.

What the cognitive investigation adds

The most solid cross-domain result in this investigation does not depend on the octave-doubling arithmetic. It is that color and music are both continuous physical dimensions that the human mind partitions into discrete categories using the same two-layer cognitive architecture: universal biological anchors, culturally-drawn boundaries. This is documented independently in color science and music cognition, and both literatures reached the same "half universal, half cultural" resolution. The mind carves both continua the same way.

What remains genuinely open

Whether the structural correspondences documented here are physically meaningful or mathematically inevitable consequences of the harmonic series is unresolved. The framework lacks a mechanism — a causal account of why the 41-octave gap should be significant rather than coincidental. The strongest version of the objection: both auditory and visual systems were optimized by independent evolutionary pressures, and finding that they correspond may be finding that two independently constrained systems happen to sit 41 octaves apart. Under this interpretation, the results tell us about body size, atmospheric physics, and the harmonic series — not about any hidden unity between the senses.

What would move this from observation to theory

A mechanism. Evidence that the 41-octave relationship between human auditory and visual ranges is not coincidental — that some evolutionary, physical, or information-theoretic process caused the two ranges to be harmonically related rather than arbitrarily separated. The solar entropy convergence is a candidate. It has not been tested. Until it is, the correspondences documented here are precisely what they are: precise, surprising, structurally coherent, and unexplained.

The visible spectrum is an almost-complete chromatic scale. The one missing note is the devil in music. Whether this is a window into the structure of reality, or the harmonic series showing up twice — as it tends to do — remains honestly open.

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PART §

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End of Modulations v4.0 Original founding document: Session 1, June 2026

Modulations

An Investigation into Color, Music, and Time. Reported as an ongoing investigation rather than a finished theory: findings, objections, and open questions are held together on purpose. The interactive edition lets you explore the color tables, scales, chords, and progressions live.

Modulations is part of Open Source, a free-to-use art & research project by Jake Galm.

Open Source · mumblejinx.github.io/opensource
Jake Galm · JakeGalm.com

The octave-doubling method itself has a documented precedent in Hans Cousto’s “Cosmic Octave” (1978/1980). This project’s contribution is the chromatic-scale result obtained from the method, not the method itself.